12	**D**	At steady state, pressure in both flasks are the same. Applying ideal gas law $P_P = P_Q$ $\frac{n_P R T_P}{V_P} = \frac{n_Q R T_Q}{V_Q}$ $\frac{M_P}{M_Q} = \frac{T_Q}{T_P}\frac{V_P}{V_Q} = 9$